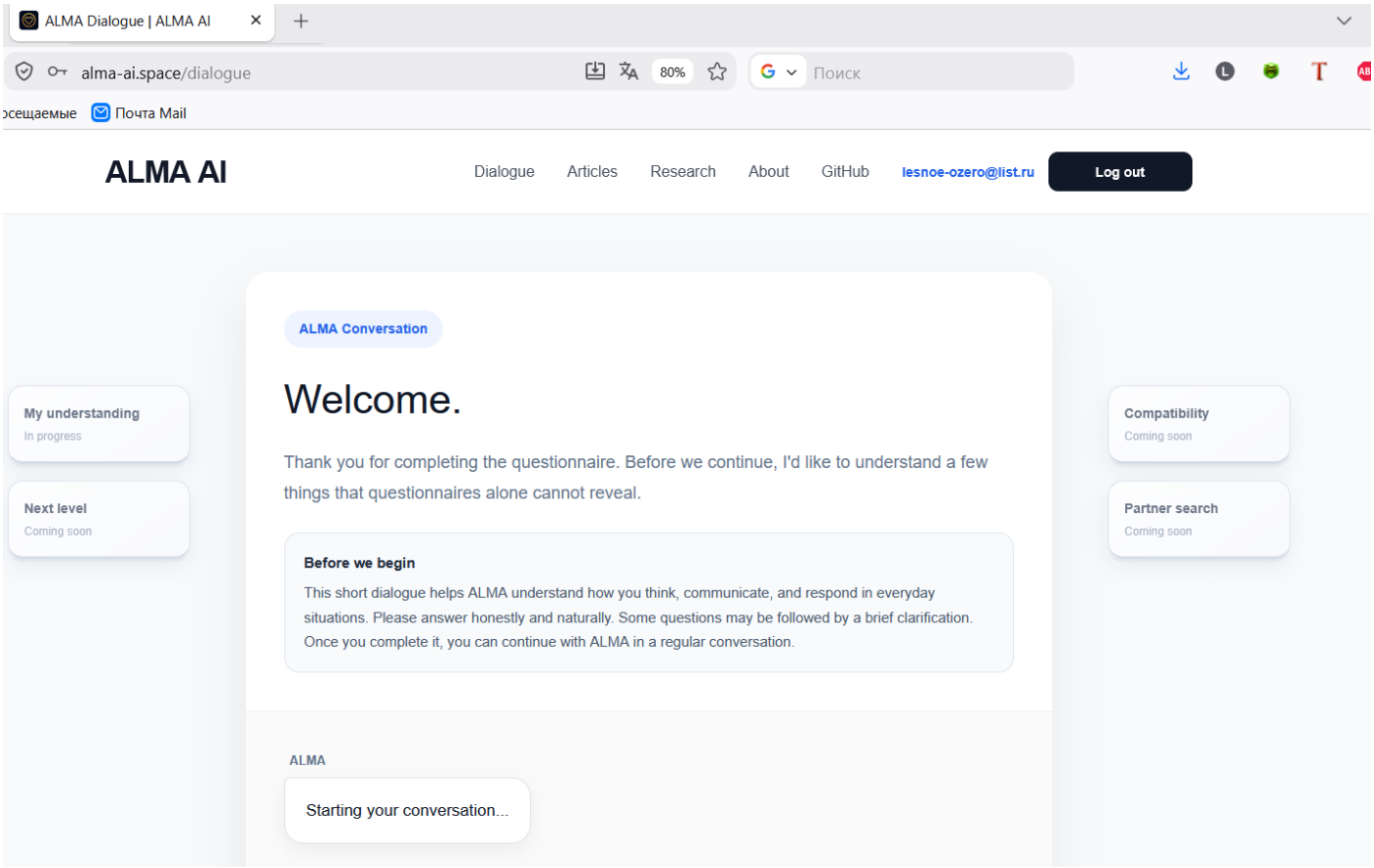
ALMA AI

ETHICAL AI FOR LONG-TERM PARTNER COMPATIBILITY

The conversation is the input.
The evolving understanding is the product.

Working research prototype • Architecture, current state & validation direction • August 2026

ALMA is a working research prototype whose core mechanism is implemented and functioning in live dialogue. The next stage is rigorous longitudinal and professional validation — not a claim of clinical validity.



Live ALMA dialogue interface — prototype stage.

01 • EXECUTIVE THESIS

From conversation to an evolving model

Most conversational AI is optimized around the next response. ALMA is designed around a different longitudinal objective: what does this interaction change about what the system understands about the person?

The LLM remains the language engine. ALMA places it inside a larger cognitive architecture that extracts observations, forms candidate hypotheses, compares new evidence with prior understanding, revises interpretations and stores the resulting person-level model for future dialogue.

A standard LLM decides what to say next. ALMA is designed to track what the current interaction changes about its evolving understanding of the person.

What the current prototype already demonstrates

- Live dialogue through the deployed ALMA web interface.
- Persistent conversation history across sessions.
- Persistent person-level memory connected to the running system.
- A modular architecture for observations, hypotheses, evidence, revision and memory.
- Document-based conversation analysis within the prototype workflow.
- A live research environment in which the system can be repeatedly observed and refined.

These capabilities establish the engineering substrate. They do not, by themselves, establish psychological validity — that is the next research layer.

Why ALMA is different from a conventional LLM

The distinction is not that ALMA uses a better language model. The distinction is what the language model sits inside of — and what the system is designed to preserve across time.

Dimension	Typical LLM interaction	ALMA approach
Primary objective	Generate a relevant response	Build and maintain psychological understanding over time
Memory	Dependent on available context and history	Persistent person-level memory
Observations	Implicitly used in generation	Explicitly extracted and structured
Hypotheses	May appear in an answer without being maintained	Generated, compared, merged and accumulated
Validation	Response quality judged turn by turn	Patterns and interpretations examined across repeated interaction
Model of the person	Reconstructed from available context	Continuously evolving cognitive representation

The LLM is still responsible for language understanding and natural-language generation. ALMA's architectural claim is that communication is embedded in a system whose primary purpose is longitudinal understanding rather than response generation alone.

The core mechanism: observation → hypothesis → revision

ALMA is not intended to turn a conversation into a fixed profile in one pass. The core principle is a feedback cycle in which new evidence can change the interpretation of earlier evidence.

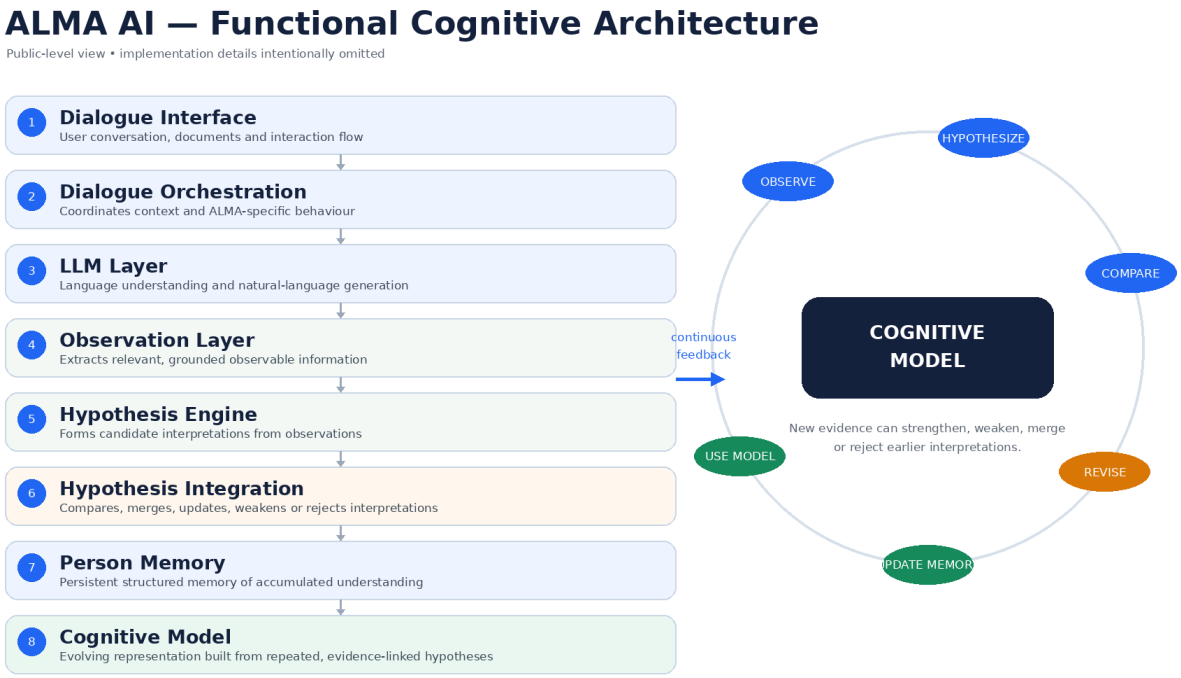
1. Observe Identify information grounded in what the person actually said or did.	**2. Form a hypothesis** Turn repeated or meaningful observations into a candidate interpretation.	**3. Compare** Compare new information with previously stored understanding.
4. Confirm / weaken / merge / reject Do not treat an interpretation as permanently true; revise it as evidence changes.	**5. Update memory** Retain relevant, structured understanding in persistent person-level memory.	**6. Use the model** Allow accumulated understanding to influence how future dialogue is interpreted and handled.

The product is not simply a memory of what a person said. It is an attempt to progressively organize evidence into a more stable — and revisable — understanding of the person.

Important validation boundary: the mechanism can be implemented and observed before the resulting psychological constructs are considered scientifically validated. ALMA therefore treats professional review and longitudinal measurement as a distinct research stage.

Functional cognitive architecture

The public architecture is shown at the functional level. It communicates the product principle while protecting proprietary implementation details.



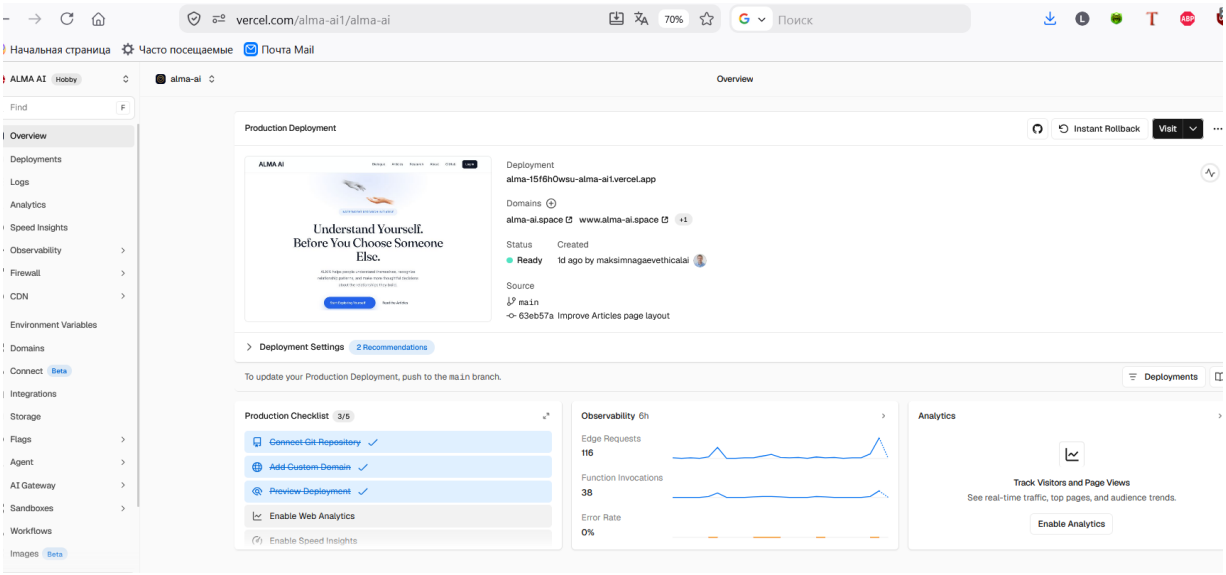
Functional architecture and feedback cycle. Proprietary implementation details intentionally omitted.

The central design feature is the feedback loop around the Cognitive Model. New observations do not merely append facts; they can change how previous evidence is interpreted. This is the architectural mechanism that makes longitudinal understanding possible.

- Dialogue interface and orchestration manage interaction flow.
- The LLM layer provides language understanding and generation.
- Observation and hypothesis components separate what was observed from what is inferred.
- Hypothesis integration and person memory preserve accumulated evidence.
- The cognitive model is the evolving representation used by subsequent dialogue.

What exists today

ALMA is not a static local demonstration. The web application is deployed, the backend is running as a web service, and persistent storage is connected to the live system.



Production deployment view for the ALMA web application.

The current prototype is best described as a working research system: the core mechanism is implemented and functioning in live interaction, while the quality, consistency and usefulness of the resulting cognitive model remain subjects of ongoing validation.

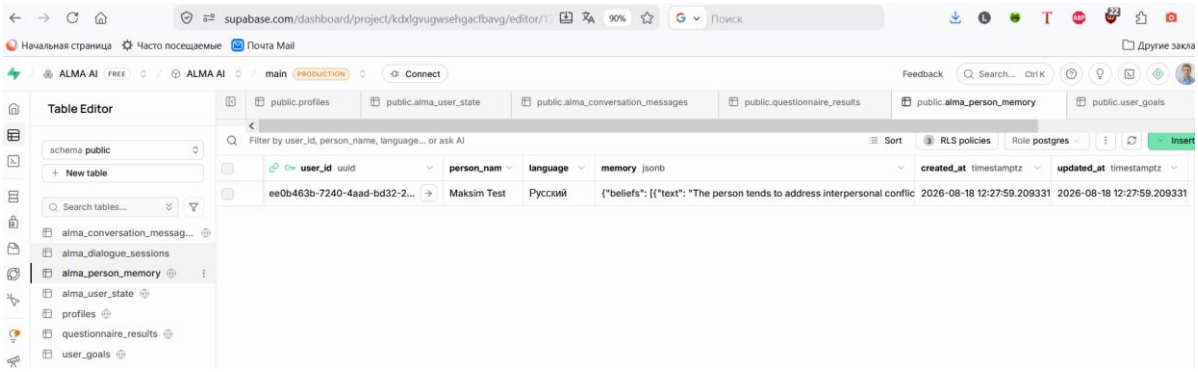
Engineering status: the architecture is operational. Research status: the psychological methodology and longitudinal performance still need to be established.

We intentionally do not present unverified scale claims here. Quantitative longitudinal metrics will be added when the testing period and measurement protocol are sufficiently mature to support them.

06 • EVIDENCE

Prototype evidence and persistence

The current evidence is strongest when read as a set of complementary artifacts: a live interface, a functional cognitive architecture, persistent person-level memory and a deployed runtime.



Supabase view showing persistent ALMA person-level memory.

Persistent storage demonstrates that the person-level model is not limited to a single browser session. The stored record is connected to the running backend and can be loaded again when a dialogue is resumed.

This is evidence of persistence and system integration — not evidence that the psychological content of the model is already valid. That distinction is central to ALMA's research posture.

What we deliberately do not use as proof

Raw IDE file trees are omitted from the partner-facing version. They demonstrate implementation activity, but they do not demonstrate that the cognitive mechanism is psychologically meaningful. The architecture diagram communicates that point more efficiently.

What remains open — and how it should be validated

The central research questions are now clearer than the engineering questions. The goal is not to claim that ALMA's interpretation of a person is automatically correct, but to determine whether the process produces representations that are coherent, revisable, useful and defensible.

Focus	Question	What to examine	Validation direction
Coherence over time	Does stored understanding become more coherent as evidence accumulates?	Longitudinal comparison of model states across repeated interactions.	Define protocol → collect longitudinal evidence → independent review
Revisability	Do hypotheses remain appropriately revisable instead of becoming fixed labels?	Track hypothesis changes when later evidence supports, weakens or contradicts earlier interpretations.	Define protocol → collect longitudinal evidence → independent review
Use in dialogue	Does accumulated understanding materially influence later dialogue?	Controlled comparisons of later responses with and without relevant model context.	Define protocol → collect longitudinal evidence → independent review
Longitudinal usefulness	Does the quality of the model improve with sustained interaction?	Predefined evaluation criteria applied across multiple sessions and time periods.	Define protocol → collect longitudinal evidence → independent review
Psychological meaningfulness	Can professional reviewers agree that observations, hypotheses and evidence are meaningful?	Structured review with psychologists / relationship researchers against explicit criteria.	Define protocol → collect longitudinal evidence → independent review

The document intentionally separates these questions from current engineering status. The prototype can be operational before its psychological methodology is validated; the research program is what must establish the latter.

The next stage is methodological, not merely technical

The next stage is not primarily about adding generic AI capability. It is about making the cognitive-model methodology more rigorous and validating it against real longitudinal interaction.

Partner type	Why now
Psychologists and relationship researchers	Define meaningful observations, hypotheses and evidence standards; challenge interpretations that are too broad, too fixed or insufficiently grounded.
Methodology and measurement experts	Design defensible evaluation criteria, longitudinal protocols and review frameworks.
AI / product / UX partners	Translate the research prototype into a scalable product without losing the core cognitive-model principle.

The engineering exists. The next challenge is to determine what counts as evidence — and how well the resulting model can stand up to it.

Near-term priorities

- Professional review of observation quality, hypothesis formulation, evidence standards and revision behavior.
- Longitudinal validation across repeated conversations and time.
- More natural integration of accumulated understanding into dialogue.
- Improved handling of deliberate topic shifts without removing user freedom.
- Stronger input capabilities, UX, privacy and production-readiness layers.

09 • FROM INSIGHT TO ACTION

Product philosophy and commercial direction

ALMA is not designed as a therapist, diagnostician or generic chatbot. Its intended role is to help a person achieve greater psychological clarity and understand relationship dynamics and compatibility.

Insight → Reflection → Action

The evolving cognitive model is the foundation that makes this progression possible. Reflection and practical action can become supporting layers around the core understanding system; therapeutic or emotional-regulation techniques remain secondary rather than the central product identity.

Commercial confidentiality

This overview intentionally describes the architecture at a public functional level. It does not disclose proprietary prompts, internal data schemas, implementation details, scoring thresholds, source code, orchestration rules or other information that could materially expose the underlying technology.

The purpose of this document is to demonstrate the technological principle, current prototype status and research opportunity — not to disclose the implementation itself.

What ALMA is ultimately trying to make possible

A relationship system that does not treat every conversation as an isolated event, and does not reduce a person to a static questionnaire result. Instead, it continuously works with evidence, uncertainty and revision to build a more useful understanding over time.

10 • CONCLUSION

A different kind of conversational system

The central technical hypothesis has reached the prototype stage and is functioning in live interaction. ALMA can maintain persistent person-level memory and is built around an architecture in which observations and hypotheses accumulate into an evolving cognitive model.

The remaining challenge is no longer whether the concept can be prototyped. The challenge is to establish — with professional psychological input and longitudinal evidence — how rigorous, useful, consistent and scalable that model can become.

The conversation is the input.
The evolving understanding is the product.

ALMA AI is a working research prototype for longitudinal psychological understanding. The engineering principle is implemented; the research opportunity is to determine how well that principle can work when subjected to rigorous evidence and professional scrutiny.

ALMA AI

https://alma-ai.space/

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